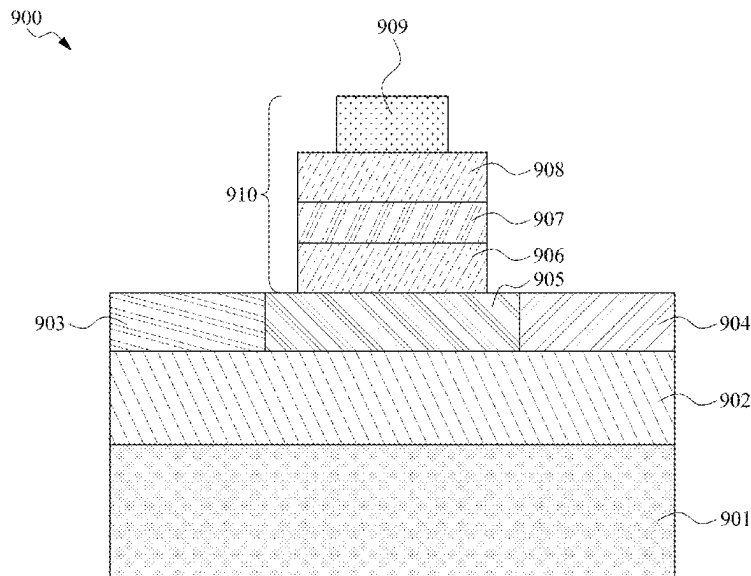
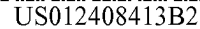




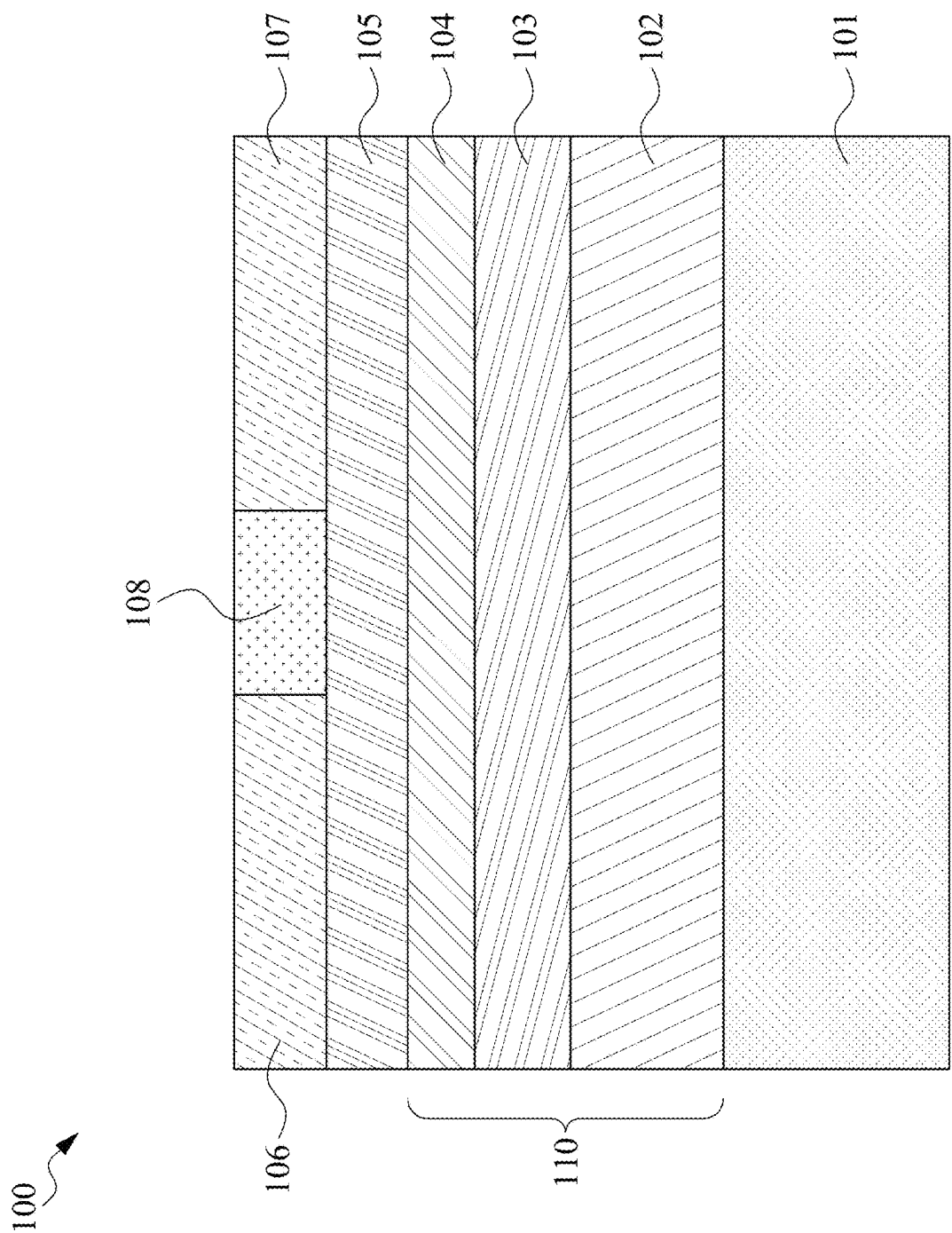


FIG. 1

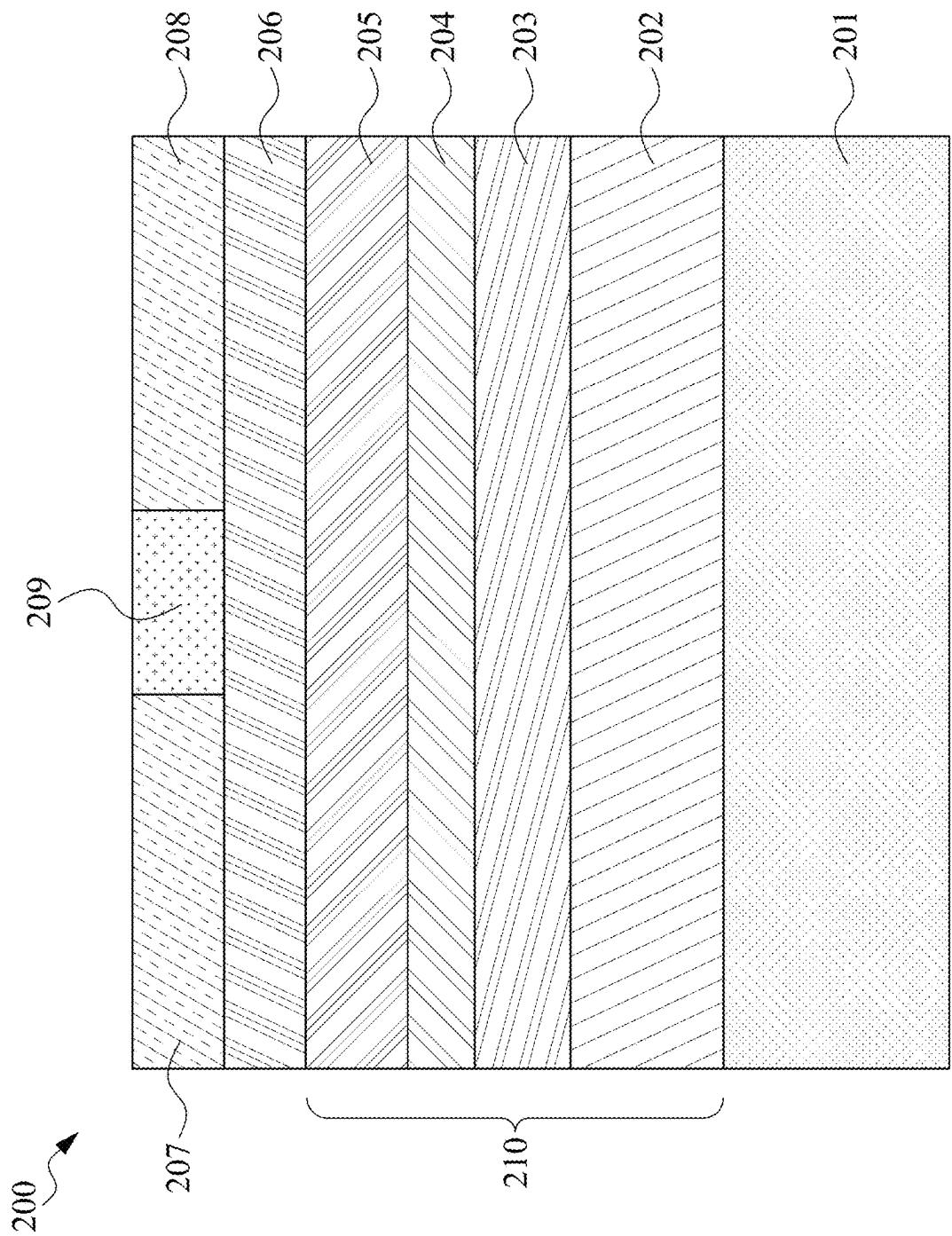


FIG. 2

300

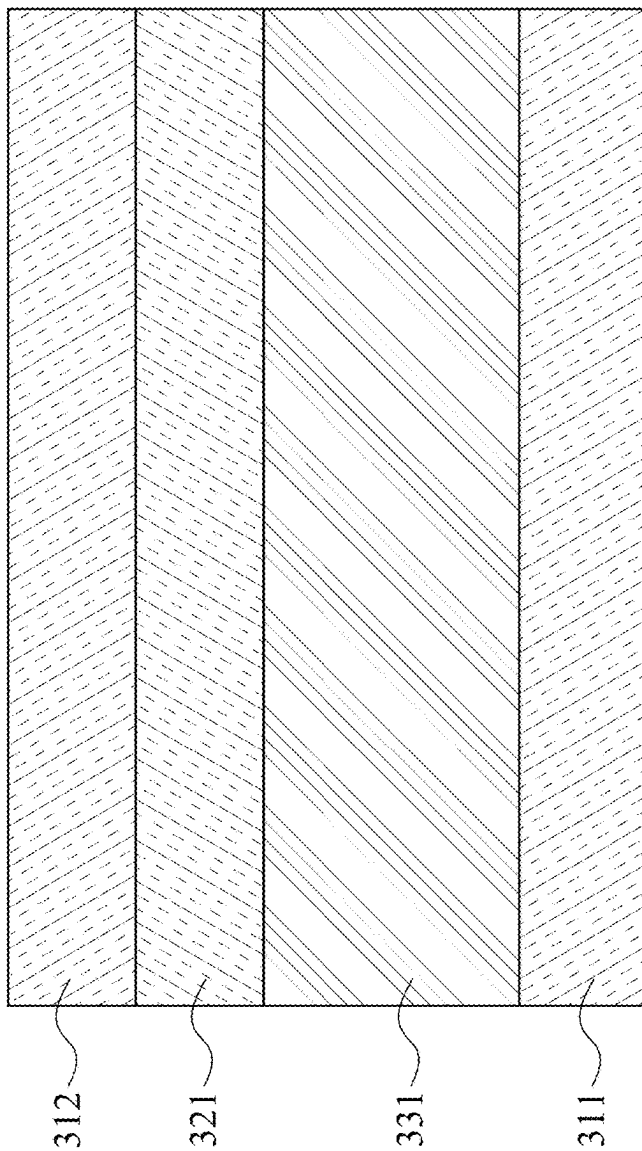


FIG. 3

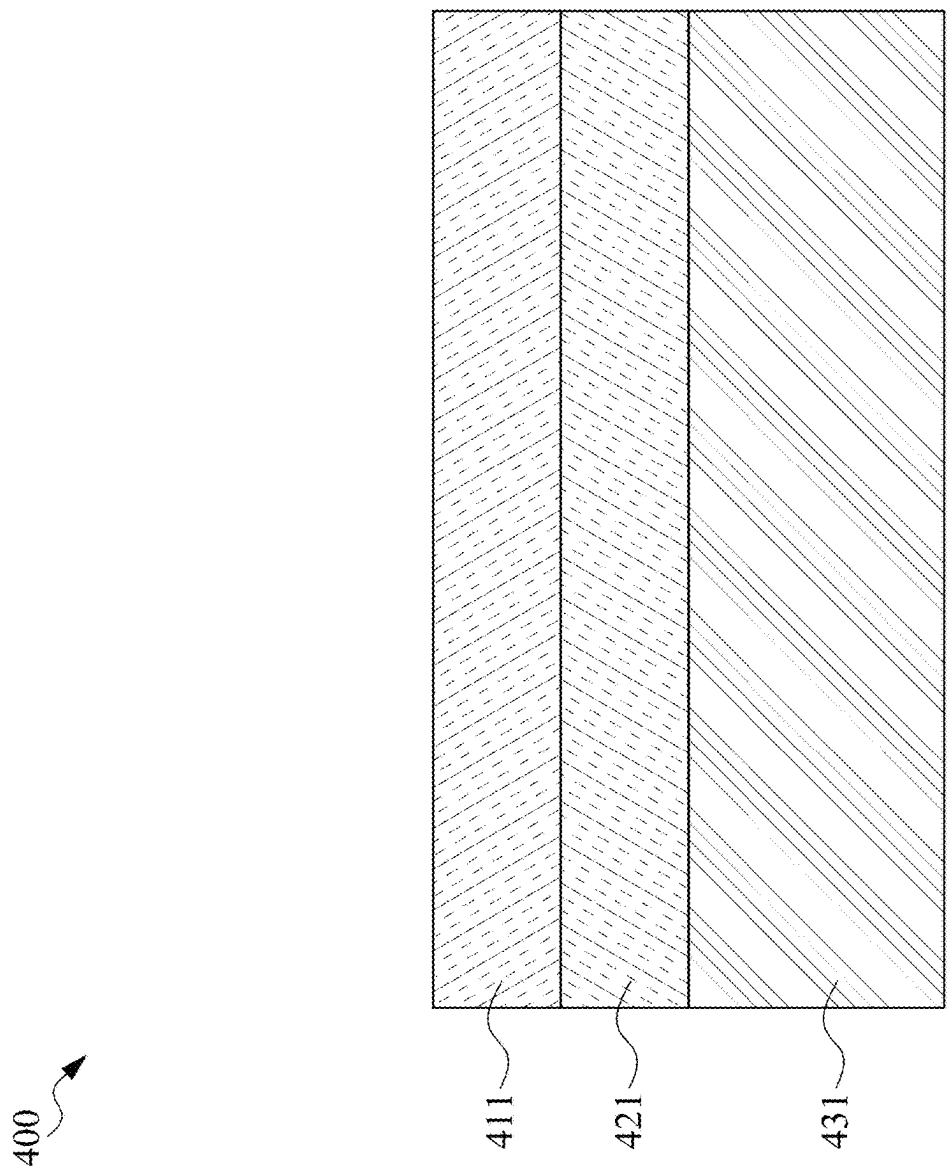


FIG. 4

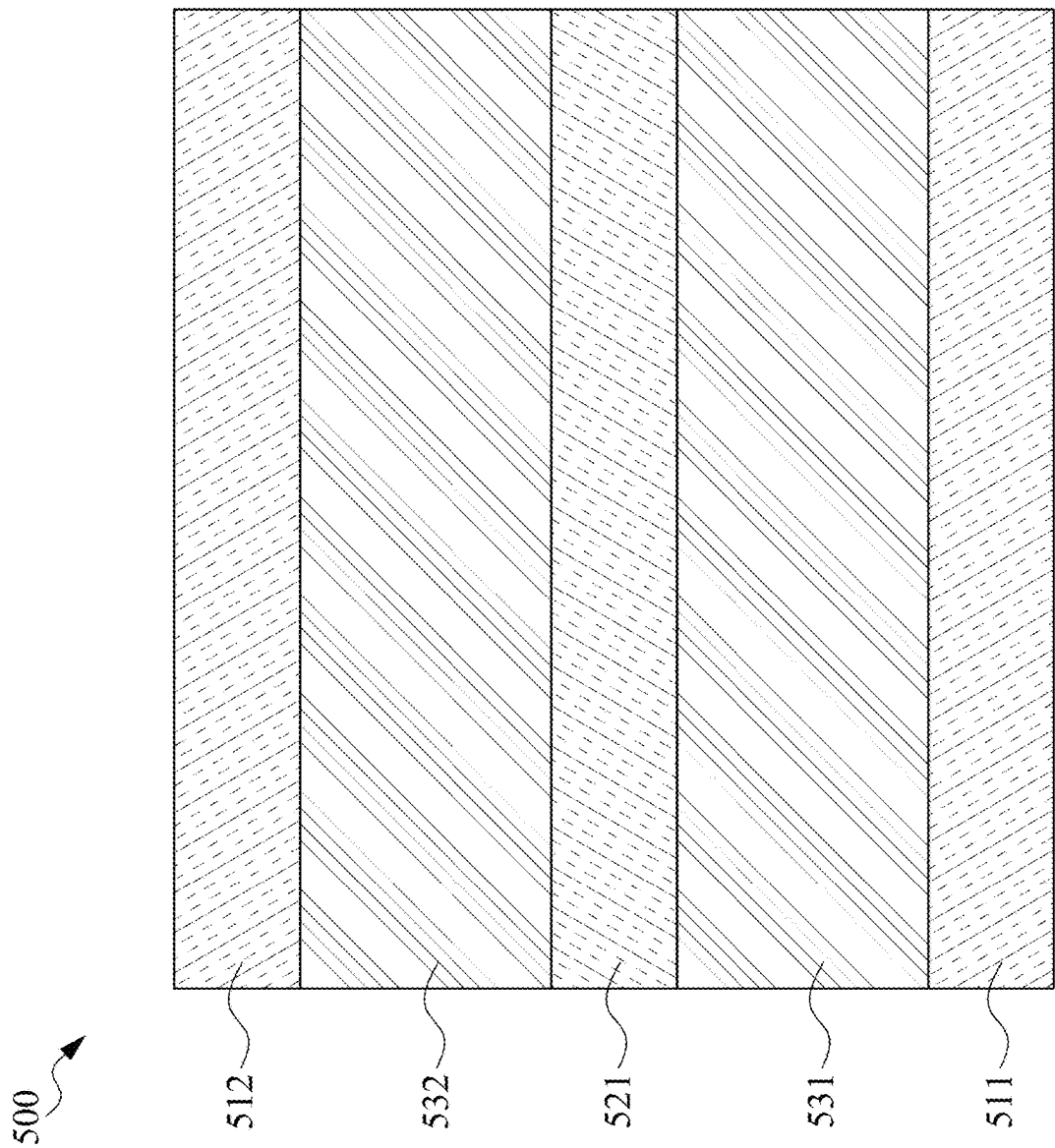


FIG. 5

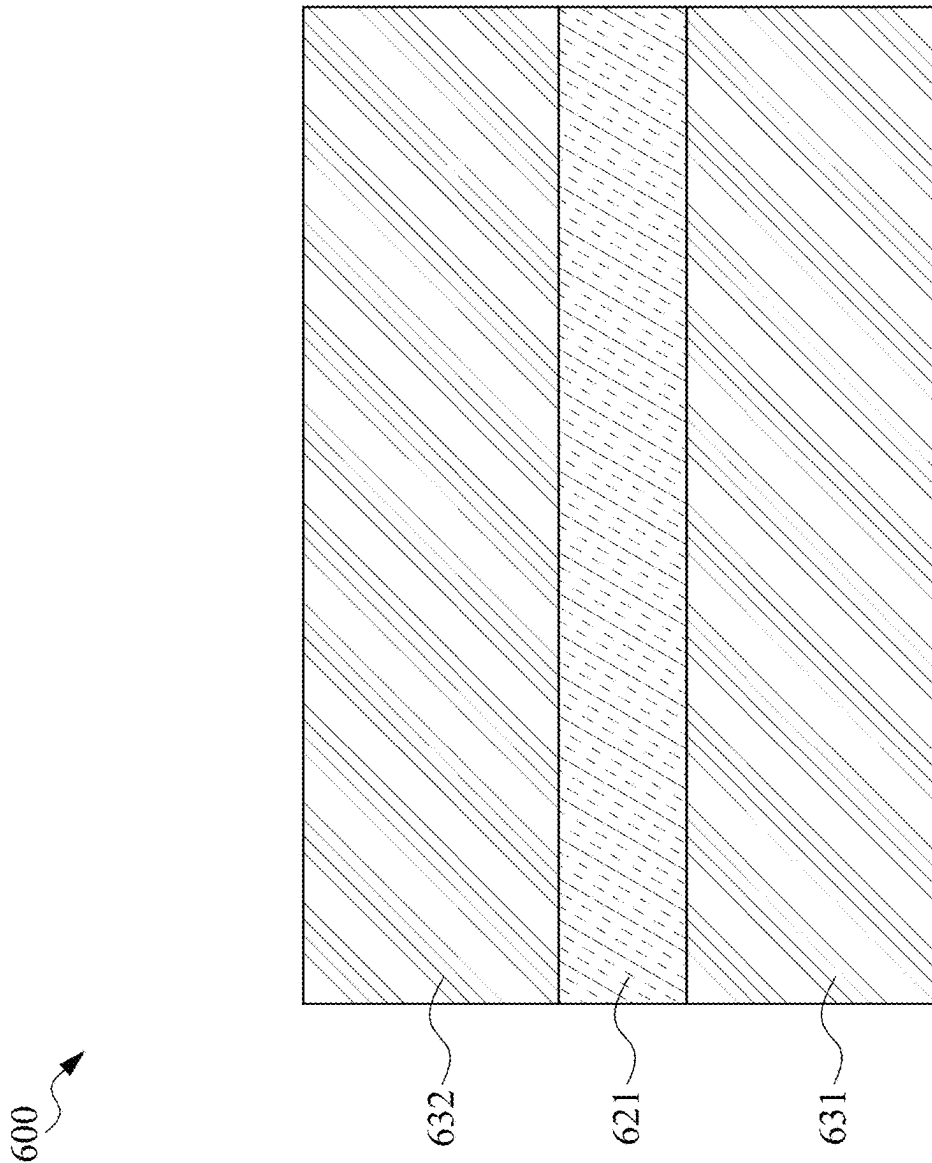


FIG. 6

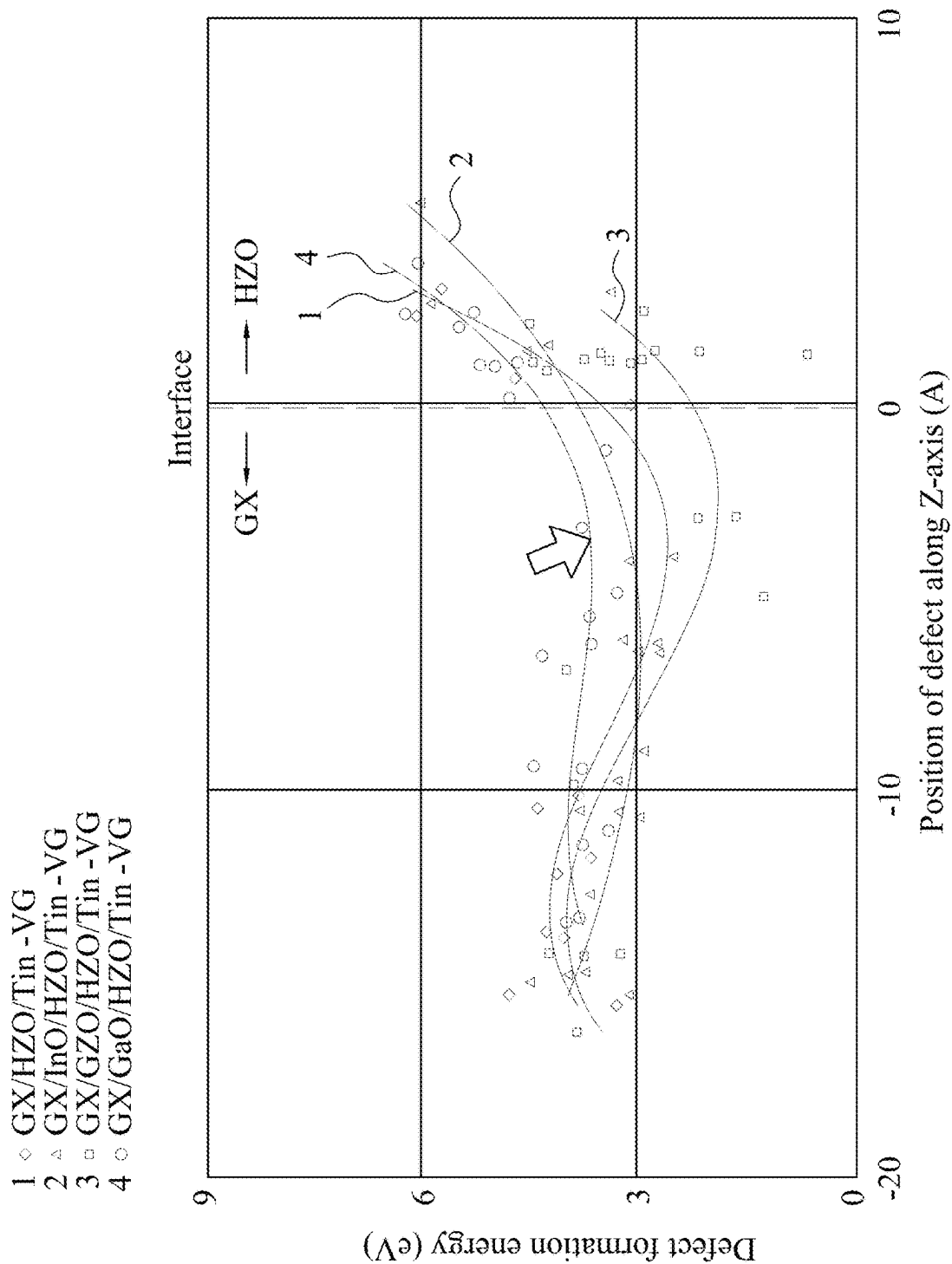


FIG. 7

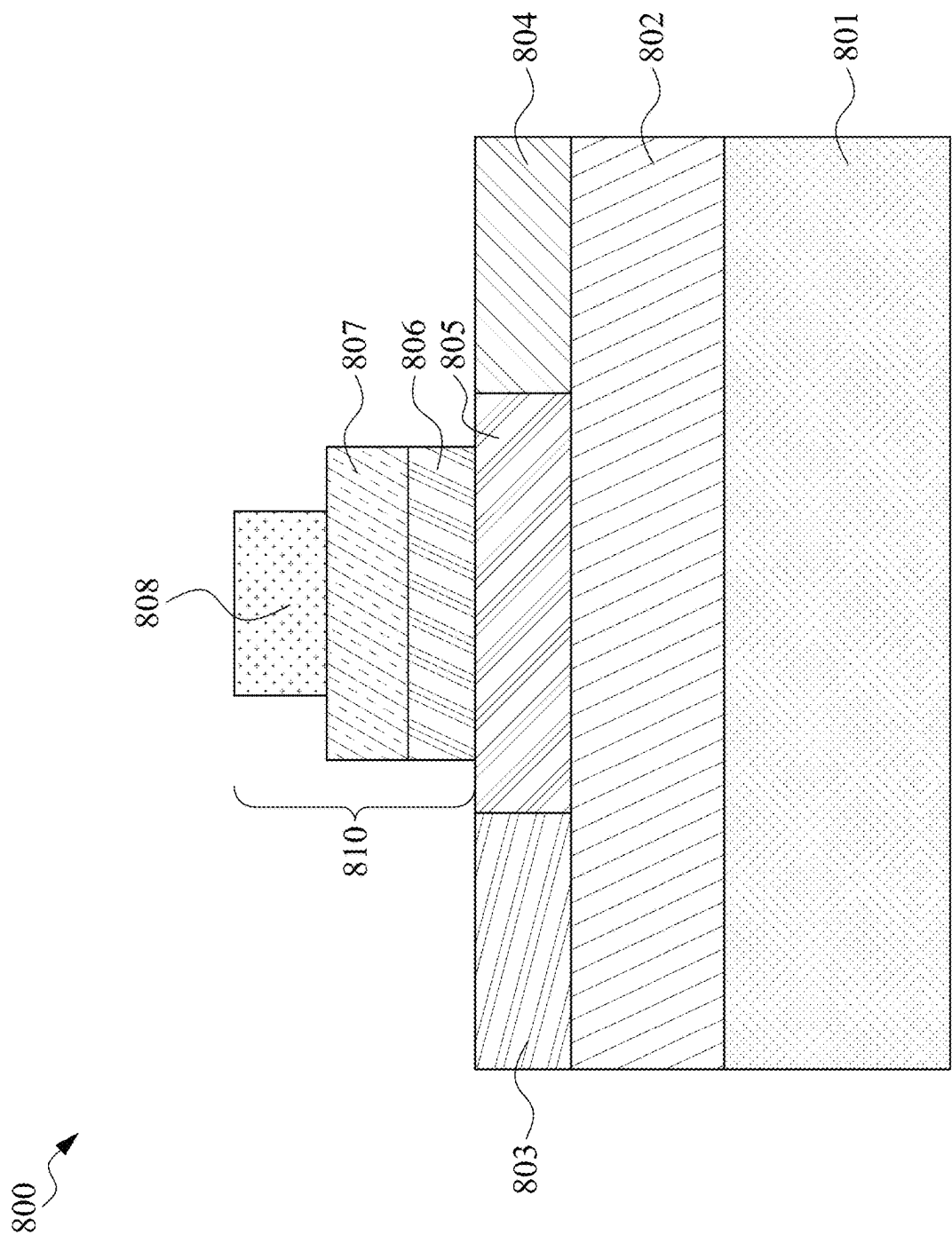


FIG. 8

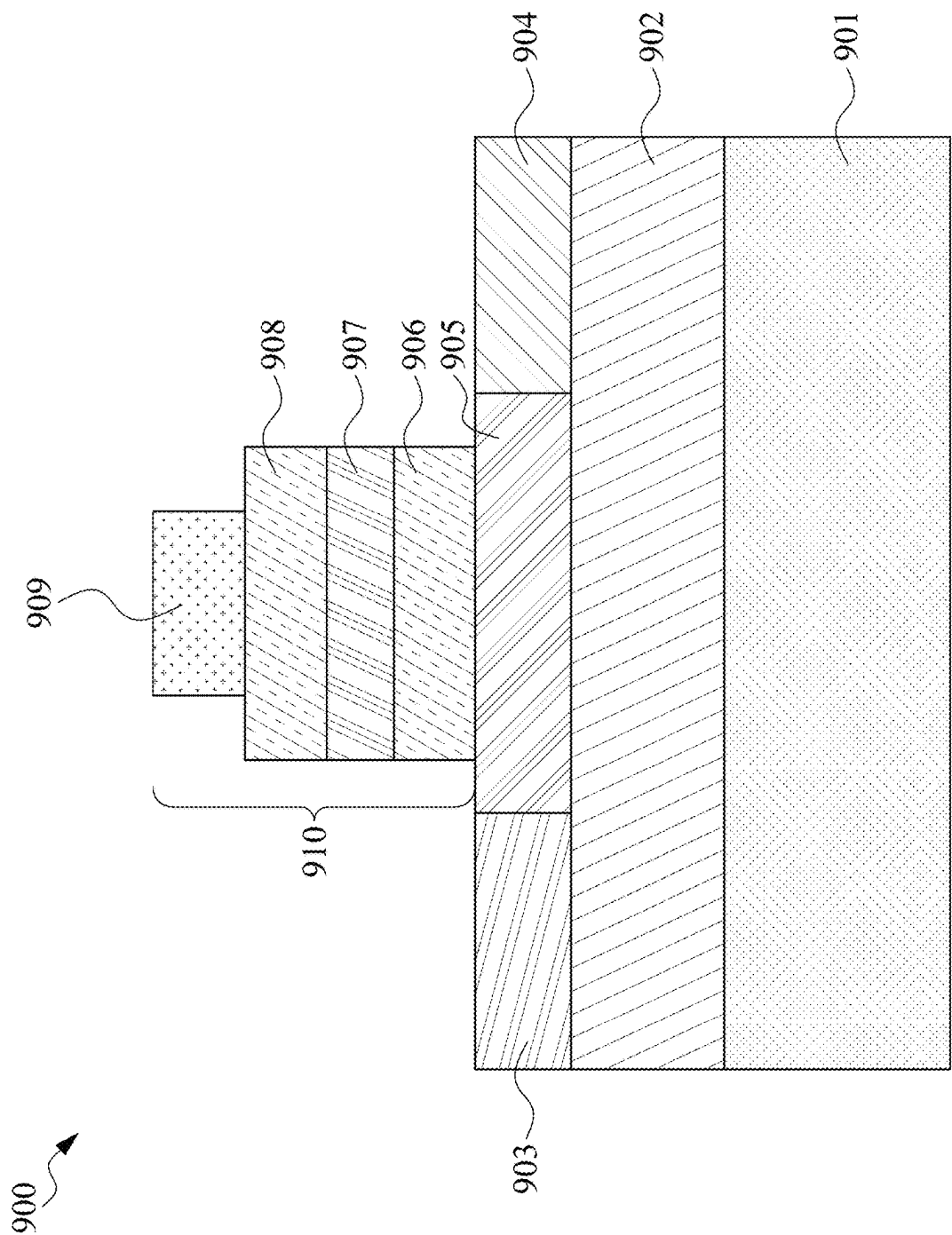


FIG. 9

SEMICONDUCTOR DEVICE, FERROELECTRIC CAPACITOR AND LAMINATED STRUCTURE

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a continuation application of U.S. non-provisional application Ser. No. 18/192,681 filed Mar. 30, 2023, which is a continuation application and claims the benefit of U.S. non-provisional application Ser. No. 17/160,013 filed Jan. 27, 2021, now U.S. Pat. No. 11,621,337B2, the disclosures of which are hereby incorporated by reference in its entirety.

BACKGROUND

Typically, integrated circuits (ICs) include a considerable amount of circuit elements formed on a given area of a chip. The circuit element is represented by a semiconductor device. For example, current advanced ICs are formed by millions of field effect transistors (FETs), which are also referred to as metal-oxide-semiconductor (MOS) transistors or MOSFETs. In general, MOSFETs may be considered as dominant semiconductor devices in modern ICs. A ferroelectric field-effect transistor (Fe FET) is a type of FET that includes a ferroelectric material sandwiched between the gate electrode and source-drain conduction region of the device. Permanent electrical field polarization in the ferroelectric causes this type of device to retain the transistor's state (on or off) in the absence of any electrical bias. FeFET based devices are used in FeFET memory that is a type of single transistor non-volatile memory. Hence, it is desirable to provide a semiconductor device with incorporated ferroelectric materials in good compatibility with standard manufacturing techniques and without deteriorated performance of fabricated semiconductor devices.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates cross-sectional views of a ferroelectric field-effect transistor (FeFET) in accordance with the embodiment of the present disclosure.

FIG. 2 illustrates cross-sectional views of a Ferroelectric field-effect transistor (FeFET) in accordance with the embodiment of the present disclosure.

FIG. 3 illustrates a structure of ferroelectric capacitor in accordance with the embodiment of the present disclosure.

FIG. 4 illustrates a structure of ferroelectric capacitor in accordance with the embodiment of the present disclosure.

FIG. 5 illustrates a structure of ferroelectric capacitor in accordance with the embodiment of the present disclosure.

FIG. 6 illustrates a structure of ferroelectric capacitor in accordance with the embodiment of the present disclosure.

FIG. 7 illustrates the simulated results of the endurance of a device in accordance with the embodiment of the present disclosure.

FIG. 8 illustrates cross-sectional views of a ferroelectric field-effect transistor (FeFET) in accordance with the embodiment of the present disclosure.

FIG. 9 illustrates cross-sectional views of a Ferroelectric field-effect transistor (FeFET) in accordance with the embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “underlying,” “below,” “lower,” “overlying,” “upper” and the like, may be used herein for case of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

In some comparative approaches, in a ferroelectric field-effect transistor (FeFET), the ferroelectric (FE) layer of a gate directly contacts with a metal electrode. Similarly, a layer of ferroelectric material in a ferroelectric capacitor is typically sandwiched between a pair of metal electrodes, forming a metal-ferroelectric-metal (MFM) structure. The ferroelectrics have a property of the spontaneous polarization and the polarization can be reversed by a suitably strong applied electric field in the opposite direction. The spontaneous polarization of the ferroelectrics implies a hysteresis effect which can be used as a memory function. Ferroelectric capacitors are indeed used to manufacture ferroelectric RAM (FeRAM) for computers and RFID cards. Typically, ferroelectric materials are used in the form of thin films to allow the field required to switch the polarization.

However, it should be paid special attention to work reliability of the interfaces between metal electrodes and ferroelectric layers when using the thin films. After repetitive electrical cycling, the switchable polarization of the ferroelectric material will decay or disappear. The decrease of switchable polarization in a ferroelectric material causes threshold voltage (VTH) shift, which can be referred to as a “fatigue” effect. The polarization fatigue may be resulted from charge injections or defects such as oxygen vacancies due to direct contact of the ferroelectric layer with the metal electrode layer. Therefore, the present disclosure provides a semiconductor device that helps to mitigate the polarization fatigue issue.

Semiconductor devices of ferroelectric field effect transistor (FeFET) are provided in accordance with various exemplary embodiments. The variations of the embodiments

are discussed. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements.

Embodiments such as those described herein provide a semiconductor device including: a substrate, a gate stack over the substrate, a channel layer over the gate stack, and a source/drain electrode disposed on sides of the gate stack. The gate stack includes a metal gate electrode, a ferroelectric layer, and a (semi) conducting oxide layer disposed between the ferroelectric layer and the metal gate electrode. In various embodiments, the gate stack includes a ferroelectric layer and a (semi) conducting oxide layer as a metal gate electrode. In various embodiments, the semiconductor device has a single gate stack, double gate stacks, or multiple gate stacks, such as fin-like FET (FinFET). In other embodiments, the ferroelectric layer may be alternatively replaced by an anti-ferroelectric layer. The use of the (semi) conducting oxide layer disposed between the ferroelectric layer and the metal gate electrode layer allows formation of FeFET devices with mitigated fatigue issue compared to FeFET devices without the semiconducting oxide layer.

In accordance with yet alternative embodiments of the present disclosure, the semiconductor device may include static random access memory (SRAM) and/or other logic circuits, passive components such as resistors, capacitors, and inductors, and active components such as P-channel field effect transistors (PFET), N-channel FET (NFET), complementary metal-oxide semiconductor (CMOS) device, bipolar transistors, high voltage transistors, high frequency transistors, other memory cells, and combinations thereof. However, further embodiments are applicable to other types of semiconductor devices or components.

FIG. 1 illustrates a cross-sectional view of a semiconductor device of a ferroelectric field-effect transistor (FeFET) device **100** in accordance with some embodiment of the present disclosure. In some embodiments, the FeFET device **100** as mentioned above adopts a bottom-gate structure, but the disclosure is not limited thereto. Referring to FIG. 1, the FeFET device **100** includes a gate stack **110** disposed over a substrate **101**. The gate stack **110** includes a metal gate electrode **102** over the substrate **101**, an oxide layer **103** over the metal gate electrode **102**, and a ferroelectric layer **104** over the oxide layer **103**. In alternative embodiments, the gate stack includes a conducting oxide layer as a gate electrode over a substrate and a ferroelectric layer over the conducting oxide layer. In some embodiments, the oxide layer **103** can be a conducting oxide layer. In some embodiments, the oxide layer **103** can be semiconducting oxide layer. The FeFET device **100** further includes a channel layer **105**, a source electrode **106** and a drain electrode **107** disposed over the gate stack **110**. The source electrode **106** and the drain electrode **107** are separated by an insulating structure **108**.

In accordance with some embodiments of the present disclosure, with reference to FIG. 1, the substrate **101** includes silicon (Si). Alternatively, the substrate **101** may include other elementary semiconductor such as germanium (Ge) in accordance with some embodiments of the present disclosure. In some embodiments, the substrate **101** additionally or alternatively includes a compound semiconductor such as silicon carbide (SiC), silicon oxide, gallium arsenic (GaAs), indium arsenide (InAs), and indium phosphide (InP). In some embodiments, the substrate **101** includes an alloy semiconductor such as silicon germanium (SiGe), silicon germanium carbide (SiGeC), gallium arsenic phosphide (GaAsP), and gallium indium phosphide (GaInP).

In accordance with some embodiments of the present disclosure, the substrate **101** may also be in the form of silicon-on-insulator (SOI). Generally, an SOI substrate includes a layer of a semiconductor material (e.g., silicon, germanium and/or the like) formed on an insulator layer. In some embodiments, the insulator layer may be, for example, a buried oxide (BOX) layer or a silicon oxide layer. In some embodiments, other substrates that may be used include multi-layered substrates, gradient substrates, hybrid orientation substrates, any combinations thereof and/or the like. In some embodiments, the substrate **101** includes various p-type doped regions and/or n-type doped regions, such as p-type wells, n-type wells, p-type source/drain features and/or n-type source/drain features, formed by a process such as ion implantation and/or diffusion. The substrate **101** may include other functional features such as a resistor, a capacitor, diode, transistors, such as field effect transistors (FETs). The substrate **101** may include lateral isolation features configured to separate various devices formed on the substrate **101**.

In some embodiments, the metal gate electrode **102** may include: Cu, Ag, Au, Al, Ti, Ta, W, Ni, Pt, Ru, Ni, their alloy, compound including nitride.

In accordance with some embodiments of the present disclosure, channel layer **105** may include Si, Ge, or an alloy thereof, III-V semiconductors, II-VI semiconductors, and a metallic oxide with a different proportion of Ga, Al, In, Sn, Cu, Zn, W, and/or Al. In accordance with some embodiments of the present disclosure, the channel layer has a thickness between 1 nm and 30 nm.

In accordance with some embodiments of the present disclosure, the source electrode **105** and the drain electrode **106** are disposed on opposite sides of the channel layer **105** and separated by an insulating structure **108**. In some embodiments, the source electrode **105** and the drain electrode **106** include metallic materials. In some embodiments, the source electrode **105** and the drain electrode **106** may be formed by using at least one metal such as aluminum (Al), neodymium (Nd), silver (Ag), chromium (Cr), titanium (Ti), tantalum (Ta), or molybdenum (Mo), or an alloy including the same, or a metal oxide conductive film such as indium tin oxide (ITO).

In accordance with some embodiments of the present disclosure, the insulating structure **108** is disposed to separate the source electrode **105** from and the drain electrode **106** on the channel layer **105**. In some embodiments, the insulating structure **108** includes silicon oxide. In some embodiments, the insulating structure **108** includes a high-k dielectric material.

In accordance with some embodiments of the present disclosure, a method for forming the FeFET device **100** may include: providing or receiving the substrate **101**; forming the gate stack **110** over the substrate **101**; forming the channel layer **105** over the gate stack **110**, and forming the source electrode **106** and the drain electrode **107** over the channel layer **105**. The method further includes forming the insulating structure **108** to separate the source electrode **106** and the drain electrode **107**. The formation of the gate stack **110** includes forming the metal gate electrode **102** over the substrate **101**, forming the oxide layer **103** over the metal gate electrode **102**, and forming the ferroelectric layer **104** over the oxide layer **103**. In alternative embodiments, the formation of the gate stack includes forming a conducting oxide layer as a gate electrode over a substrate and forming a ferroelectric layer over the conducting oxide layer. The oxide layer **103** is disposed between the metal gate electrode **102** and the ferroelectric layer **104**. It should be noted that

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the metal gate electrode **102** and the ferroelectric layer **104** are separated from each other by the oxide layer **103**, as shown in FIG. 1.

In accordance with some embodiments of the present disclosure, the metal gate electrode **102** is formed over the substrate **101** and contacts with the (semi) conducting oxide **103**. In some embodiments, the metal gate electrode **102** includes a conductive layer having a proper work function. For example, a p-type work function metal (p-metal) includes TiN, TaN and/or a carbon-doped metal nitride such as TaCN, whereas an n-type work function metal (n-metal) includes Ta, TiAl, and/or TiAlN. In accordance with some embodiments of the present disclosure, the metal gate electrode **102** includes other conductive materials, such as silver, aluminum, copper, tungsten, nickel, metal alloys, metal silicide, other suitable materials, or combinations thereof. The metal gate electrode layer **102** may be made of a conductive material, such as metals, metal nitrides, or combinations thereof. In some embodiments, the metal gate electrode **102** can include a single-layered structure. In some alternative embodiments, the metal gate electrode **102** can include a multi-layered structure. In some embodiments, the metal gate electrode may be not necessary in the presence of a conducting oxide as a gate electrode over the substrate.

In accordance with some embodiments of the present disclosure, whether in the top-gate approach or the bottom-gate approach, the oxide layer **103** is disposed between the metal gate electrode **102** and the ferroelectric layer **104**. In accordance with some embodiments of the present disclosure, the metal gate electrode **102** and the ferroelectric layer **104** are separated from each other by the oxide layer **103**. In some embodiments, the oxide layer can be a conducting oxide layer. In some embodiments, the oxide layer can be semiconducting oxide layer. The oxide layer **103** can mitigate the ferroelectric fatigue of the FeFET semiconductor device by reducing a direct contact with the metal gate electrode **102**, which may result in a decrease of switchable polarization in the ferroelectric layer **104**.

In some embodiments, the oxide layer **103** may include a semiconductor oxide materials such as, for example but not limited thereto, gallium oxide (GaO), zinc oxide (ZnO), indium oxide (InO), antimony oxide (SbO), copper oxide (CuO), indium antimony oxide (InSbO), aluminum zinc oxide (AlZnO), molybdenum disulfide (MoS₂), cadmium oxide (CdO), indium (III) oxide (In₂O₃), gallium (III) oxide (Ga₂O₃), tin (IV) oxide (SnO₂), a two-component metal oxide such as an In—Zn-based oxide, a Sn—Zn-based oxide, an Al—Zn-based oxide, a Zn—Mg-based oxide, a Sn—Mg-based oxide, an In—Mg-based oxide, or an In—Ga-based oxide; a three-component metal oxide such as an In—Ga—Zn-based oxide (also referred to as InGaZnO or IGZO), an In—Al—Zn-based oxide, an In—Sn—Zn-based oxide, a Sn—Ga—Zn-based oxide, an Al—Ga—Zn-based oxide, a Sn—Al—Zn-based oxide, an In—Hf—Zn-based oxide, an In—La—Zn-based oxide, an In—Ce—Zn-based oxide, an In—Pr—Zn-based oxide, an In—Nd—Zn-based oxide, an In—Sm—Zn-based oxide, an In—Eu—Zn-based oxide, an In—Gd—Zn-based oxide, an In—Tb—Zn-based oxide, an In—Dy—Zn-based oxide, an In—Ho—Zn-based oxide, an In—Er—Zn-based oxide, an In—Tm—Zn-based oxide, an In—Yb—Zn-based oxide, or an In—Lu—Zn-based oxide; or a four-component metal oxide such as an In—Sn—Ga—Zn-based oxide, an In—Hf—Ga—Zn-based oxide, an In—Al—Ga—Zn-based oxide, an In—Sn—Al—Zn-based oxide, an In—Sn—Hf—Zn-based oxide, or an In—Hf—Al—Zn-based oxide can be used. In some embodi-

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ments, the oxide layer **103** includes semiconducting oxides, such as SrRuO₃, InGaZnO or LaSrMnO.

In accordance with an embodiment of the present disclosure, the oxide layer **103** is an IGZO layer which contains at least four kinds of elements of indium, gallium, zinc, and oxygen, and has a composition ratio (atomic percentage) of indium as twice or more as a composition ratio of gallium and a composition ratio of zinc. The IGZO layer can be formed by a sputtering or atomic layer deposition (ALD) method with an oxide target having a composition ratio of indium:gallium:zinc=3:1:2, 1:1:1, or other ratios.

In accordance with some embodiments of the present disclosure, the oxide layer **103** is non-single-crystal and may be either amorphous or polycrystalline. Further, the oxide layer **103** may have either an amorphous structure including a portion having crystallinity or a non-amorphous structure. In some embodiments, when the oxide layer **103** is in an amorphous state, a flat surface can be obtained with relative ease, so that when a transistor is manufactured with the use of the oxide layer **103**, interface scattering can be reduced, and relatively high mobility can be obtained with relative ease.

In accordance with some embodiments of the present disclosure, in the oxide layer **103** having crystallinity, defects in the bulk can be further reduced and when a surface flatness is improved, mobility higher than that of an oxide layer **103** in an amorphous state can be obtained. In order to improve the surface flatness, the oxide layer **103** is preferably formed over a flat surface. Specifically, the oxide layer **103** may be formed over a surface with the average surface roughness (Ra) of less than or equal to 1 nm, preferably less than or equal to 0.3 nm, more preferably less than or equal to 0.1 nm. The thickness of each of the oxide layer **103** is preferably 2 nm to 50 nm.

In accordance with some embodiments of the present disclosure, the ferroelectric layer **104** is disposed between the oxide layer **103** and the channel layer **105**. In accordance with an embodiment of the present disclosure, at least a portion of the ferroelectric layer **104** is disposed between the oxide layer **103** and the metal gate electrode **102**. The ferroelectric layer **104** includes electric dipoles and can be electrically charged to form a charged layer with fixed charge. The adjacent oxide **103** can stabilize oxygen vacancies of the ferroelectric layer **104** and can reduce charge injections from the metal gate electrode **102** so as to mitigate the ferroelectric fatigue of the FeFET semiconductor device in accordance with the present disclosure.

In accordance with some embodiments of the present disclosure, the ferroelectric layer **104**, as-deposited without being annealed, may, or may not, have the ferroelectric property. However, it is still referred to as a ferroelectric layer since the ferroelectric property will be achieved in subsequent processes. In some embodiments, the ferroelectric layer **104** has a crystalline structure. In some embodiments, the ferroelectric layer **104** has a thickness between approximately 5 nm and approximately 20 nm. The exemplary materials of ferroelectric layer **104** include HfSiO_x, HfZrO_x, Al₂O₃, TiO₂, LaO_x, BaSrTiO_x (BST), PbZr_xTi_yO_z (PZT), BiFeO₃ (BFO) or (PbLa)(ZrTi)O₃ (PLZT), or the like, wherein value x, y and z are independently greater than zero and smaller than 1. The ferroelectric layer **104** may be formed using physical vapor deposition (PVD) or atomic layer deposition, which may be formed at a wafer temperature between approximately 25° C. and approximately 400° C.

FIG. 2 illustrates a cross-sectional view of a semiconductor device of a ferroelectric field-effect transistor (FeFET)

device **200** in accordance with some embodiment of the present disclosure. In some embodiments, the FeFET device **200** as mentioned above adopts a bottom-gate structure, but the disclosure is not limited thereto. Referring to FIG. 2, the FeFET device **200** includes a gate stack **210** disposed over a substrate **201**. In some embodiments, the gate stack **210** includes a metal gate electrode **202** over the substrate **201**, an oxide **203** over the metal gate electrode **202**, a ferroelectric layer **204** over the oxide layer **203** and a high-k dielectric layer **205** over the ferroelectric layer **204**. In some embodiments, the oxide layer **203** can be a conducting oxide layer. In alternative embodiments, the gate stack includes a conducting oxide as a gate electrode over the substrate, a ferroelectric layer over the conducting oxide layer and a high-k dielectric layer over the ferroelectric layer. In some embodiments, the oxide layer **203** can be semiconducting oxide layer. The FeFET device **200** further includes a channel layer **206**, a source electrode **207** and a drain electrode **208** disposed on the sides of the gate stack **210**. The source electrode **207** and the drain electrode **208** are separated by an insulating structure **209**. In some embodiments, the insulating structure **209** includes silicon oxide. In some embodiments, the insulating structure **209** includes a high-k dielectric material.

In accordance with some embodiments of the present disclosure, the substrate **201**, the metal gate electrode **202**, the semiconducting oxide **203**, the ferroelectric layer **204**, the source electrode **207** and the drain electrode **208** may respectively have the same materials as the substrate **101**, the metal gate electrode **102**, the oxide **103**, the ferroelectric layer **104**, the source electrode **106** and the drain electrode **107** as described above; therefore, those details are omitted in the interest of brevity.

In accordance with some embodiments of the present disclosure, the high-k dielectric layer **205** may include a high-k dielectric material, which is a dielectric material having the dielectric constant higher than that of thermal silicon oxide, approximately 3.9. The high-k dielectric material can be formed by a suitable process such as atomic layer deposition (ALD). Other methods to form the high-k dielectric **205** include metal-organic chemical vapour deposition (MOCVD), Plasma Enhance Chemical Vapor Deposition (PECVD), physical vapor deposition (PVD), UV-Ozone Oxidation or molecular beam epitaxy (MBE).

In some embodiments, some of the candidate materials (such as HfO_2 , HfSiO_x , HfZrO_x , Al_2O_3 , TiO_2 , and LaO_x) of the ferroelectric materials include the same elements as some high-k dielectric materials. Alternatively, the ferroelectric materials may have different properties than high-k dielectric materials. For example, the ferroelectric materials may have a resistivity lower than the respective high-k dielectric materials that contain the same type of elements.

In accordance with some embodiments of the present disclosure, the semiconductor device is directed to a ferroelectric capacitor.

FIG. 3 illustrates an embodiment of a ferroelectric capacitor **300** in accordance with the present disclosure. The ferroelectric capacitor **300** has a first electrode layer **311**, a second electrode layer **312**, an oxide layer **331**, and a ferroelectric layer **321**. As mentioned above, the oxide layer **331** can be a conducting oxide layer **331** or a semiconductor oxide layer **331**. The ferroelectric layer **321** is disposed between the first electrode layer **311** and the second electrode layer **312**, while the semiconducting oxide layer **331** is further disposed between the first electrode layer **311** and the ferroelectric layer **321**. In accordance with yet alternative embodiments of the present disclosure, the semiconducting

oxide layer **331** can be disposed between the second electrode layer **312** and the ferroelectric layer **321**, though not shown. In such embodiments, the ferroelectric layer **312** is separated from at least the first electrode layer **311** or the second electrode layer **312**.

FIG. 4 illustrates an embodiment of a ferroelectric capacitor **400** in accordance with the present disclosure. In some embodiment, a first electrode layer has the same material as a semiconducting oxide layer to form a semiconducting oxide electrode layer **431**. In such embodiment, the ferroelectric capacitor **400** has the semiconducting oxide electrode layer **431**, a second electrode layer **411**, and a ferroelectric layer **421**. Further, the ferroelectric layer **421** is disposed between the semiconducting oxide electrode layer **431** and the second electrode layer **411**.

FIG. 5 illustrates an embodiment of a ferroelectric capacitor **500** in accordance with the present disclosure. The ferroelectric capacitor has a first electrode layer **511**, a second electrode layer **512**, a first oxide layer **531**, a second oxide layer **532** and a ferroelectric layer **521**. The first oxide layer **531** and the second oxide layer **532** can be a conducting oxide layer or a semiconductor oxide layer. In some embodiments, the first semiconducting oxide layer **531** and the second semiconducting oxide layer **532** can include a same material, but the disclosure is not limited thereto. It should be noted that the first semiconducting oxide layer **531** is disposed between the first electrode layer **511** and the ferroelectric layer **521**, while the second semiconducting oxide layer **532** is disposed between the second electrode layer **512** and the ferroelectric **521**. Accordingly, the ferroelectric layer **512** is separated from both of the first and second electrode layers **511** and **512** by the first and second semiconducting oxide layers **531** and **532**.

FIG. 6 illustrates an embodiment of a ferroelectric capacitor **600** in accordance with the present disclosure. In the embodiment, a first electrode layer has the same material as a semiconducting oxide layer to form a first semiconducting oxide electrode layer **631**, and a second electrode layer has the same material as a second semiconducting oxide layer to form a second semiconducting oxide electrode layer **632**. In such embodiment, the ferroelectric capacitor **600** has the first semiconducting oxide electrode layer **631**, the second semiconducting oxide electrode layer **632**, and a ferroelectric layer **621**. The ferroelectric layer **621** is disposed between the first semiconducting oxide electrode layer **631** and the second semiconducting oxide electrode layer **632**.

FIG. 7 shows that a device in accordance with the embodiment of the present disclosure may be further improved in terms of endurance. The result shown in FIG. 7 is obtained on the basis of theoretical calculation of the structures by first-principle density-functional simulation, and it shows that a device according to the embodiment of the present disclosure can be further improved by an optimized oxide/ferroelectric interface, which prevents defect formation around the interface to improve the device endurance. In the structures, TiN is used as the metal, HfZrO_2 is used as the ferroelectric material, and InGaZnO_4 (GX) is used as the oxide, and the term (such as GaO, InO and GZO) between GX and HZO means a dominant component of GX near the interface. In principle, defect formation energy (y-axis) is associated with the possibility of defect formation, that is, the higher formation energy is, the lower possibility of defect formation is. In other words, if the defect formation energy is low, defects are easier to form and gather around the interface. The x-axis of FIG. 7 means a distance from the material interface, and the zero point of the x-axis (shown as a vertical dash line) means the interface

between GX and HZO materials. As can be seen from FIG. 7, compared to the non-optimized interface of the GX/HZO/TiN structure (line 1) and the interface of the GX/InO/HZO/TiN or GX/GZO/HZO/TiN structures (lines 2 and 3, respectively), the interface of the GX/GaO/HZO/TiN structure (line 4) exhibits a relatively higher defect formation energy around the interface. This means that defects issue in the GX/GaO/HZO/TiN structure is mitigated. In some embodiments, it is found that even if the defects are formed, the defects are distributed evenly in the device rather than being gathered near the interface.

In accordance with some embodiments of the present disclosure, the FeFET device adopts a top-gate structure, but the disclosure is not limited thereto. FIG. 8 illustrates an embodiment of a FeFET device 800 in accordance with the present disclosure, which adopts a top-gate structure. In the embodiment, the top-gate structure includes a substrate 801, a buffer layer 802, a source electrode 803 and a drain electrode 804 disposed over the buffer layer 802, a channel region 805 disposed between the source and drain electrodes, and a gate stack 810 over the source and drain electrodes 803 and 804. The gate stack 810 includes a ferroelectric layer 806 over the source electrode 803, the drain electrode 804 and the channel region 805, a metal electrode 808 and a semiconducting oxide layer 807 between the ferroelectric layer 806 and the metal gate electrode 808. In various embodiments, the gate stack 810 includes a ferroelectric layer 806 over the source and drain electrodes 803 and 804, and a semiconducting oxide layer serving as a gate electrode.

FIG. 9 illustrates an alternative embodiment of a FeFET device 900 in accordance with the present disclosure, which adopts a top-gate structure. In the embodiment, the top-gate structure includes a substrate 901, a buffer layer 902, a source electrode 903 and a drain electrode 904 disposed over the buffer layer 902, a channel region 905 disposed between the source and drain electrodes, and a gate stack 910 over the source and drain electrodes 903 and 904. The gate stack 910 includes a semiconducting oxide layer 906 over the source and drain electrodes 903 and 904 and the channel region 905, a ferroelectric layer 907 over the semiconducting oxide layer 906, a semiconducting oxide layer 908 over the ferroelectric layer 907 and a metal gate electrode 909. In various embodiments, the gate stack 910 includes a semiconducting oxide layer 906, a ferroelectric layer 907 over the semiconducting oxide layer 906, and a semiconducting oxide layer 908 serving as a gate electrode over the ferroelectric layer 907.

The buffer layer 802/902, in some embodiments, is epitaxially grown on the substrate 801/901. The epitaxial growth can use Metal-Organic Chemical Vapor Deposition (MOCVD), Molecular Beam Epitaxy (MBE), Liquid Phase Epitaxy (LPE), Vapor Phase Epitaxy (VPE), Ultra-High-Vacuum Chemical Vapor Deposition (UHVCVD), the like, or a combination thereof. The buffer layer can be a group IV material, such as Si, Ge, SiGe, SiGeSn, or the like; a group III-group V compound material, such as GaAs, GaP, GaAsP, AlInAs, AlGaAs, GaInAs, InAs, GaInP, InP, InSb, GaInAsP or the like; or the like.

In some embodiments of the FeFET device adopting a top-gate structure, the formation of the source structure and the drain structure includes applying, exposing, and developing a photoresist layer. The desirable structure for forming source structure and the drain structure is exposed through the photoresist layer. In some embodiments, an implantation is then performed to implant an n-type impurity, such as phosphorous, arsenic, antimony, or combinations thereof to

form source structure and the drain structure. In some alternative embodiments, an implantation is performed to implant a p-type impurity, such as boron, indium, or combinations thereof to form drain structure and the drain structure.

In accordance with some embodiments of the present disclosure, a method including: providing a substrate; forming a buffer layer over the substrate, forming a source electrode and a drain electrode over the buffer layer; forming a gate stack over the source and drain electrodes, is provided. The formation of the gate stack includes forming a ferroelectric layer over the source and drain electrodes, forming a semiconducting oxide layer over the ferroelectric layer, and forming a metal gate electrode over the semiconducting oxide layer. In alternative embodiments, the formation of the gate stack further includes forming a semiconducting oxide layer between the ferroelectric layer and the source and drain electrodes. In yet alternative embodiments, the formation of the gate stack includes forming a ferroelectric layer over the source and drain electrodes, and forming a semiconducting oxide layer as a gate electrode over a ferroelectric layer. It should be noted that the metal gate electrode and the ferroelectric layer are separated from each other by the semiconducting oxide layer.

The disclosed technique is directed to a laminated structure including a ferroelectric layer, an electrode layer over the ferroelectric layer and a semiconducting oxide layer disclosed between the ferroelectric layer and the electrode layer.

In some embodiments according to the present disclosure, the laminated structure further includes a channel layer and a source/drain electrode, wherein the channel layer is between the source/drain electrode and the ferroelectric layer. In some embodiments according to the present disclosure, the source/drain electrode is disposed over the ferroelectric layer and the channel layer. In accordance with yet alternative embodiments of the present disclosure, the source/drain electrode is disposed under the ferroelectric layer, and in a substrate.

In some embodiments according to the present disclosure, the laminated structure further includes a second electrode layer, wherein the ferroelectric layer and the semiconducting oxide layer are between the first electrode layer and the second electrode layer.

In an embodiment, the laminated structure is part of a gate electrode of a transistor device. The transistor device has a silicon channel formed on a silicon-on-insulator substrate, and a gate structure at least partially over the silicon channel. The gate structure may include a high-k gate dielectric layer, e.g., of HfO₂, and a laminated structure over the high-k gate dielectric layer. The laminated structure includes a vertical stack of TiN, ferroelectric ZrO₂, TiN, and Pt. In combination with a nanoscale silicon strip configured as a junctionless transistor, the laminated gate stack enables a subthreshold swing ("SS") less than 60 mV/dec operated at a relatively large V_{ds} of approximately IV.

The thickness of the ferroelectric layer as mentioned above can be varied to modify the parameters of one or more of the resistance component, inductance component, or capacitance component of the laminated structure. For example, a thicker ferroelectric layer in the laminated structure brings approximately a larger resistance value of the resistance component and a lower capacitance value of the capacitance component. The thickness of the ferroelectric layer also affects the multi-domain characteristics of the laminated structure. In an example, an area size of the Pt metal layer affects the capacitance value of the laminated.

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The capacitance value of laminated structure increases with enlarged area size of the Pt metal layer.

With the proper C, L, or R values, a laminated structure as described here may be used for various circuit applications, like LC resonant circuit, band-pass circuit, band-stop circuit, low-pass filter, high-pass filter, oscillators, or negative capacitors.

In the present disclosure, a ferroelectric-based semiconductor device, capacitor or laminated structure with improvement is disclosed. In accordance with the present disclosure, a semiconducting oxide layer is disposed adjacent to a ferroelectric layer so as to preventing from charges injected from a metal electrode, thereby mitigating ferroelectric fatigue. In addition, the semiconducting oxide can prevent from trapping of injected charges and stabilize oxygen vacancies in the ferroelectric layer. Hence, ferroelectric-based semiconductor device, capacitor or laminated structure can significantly improve the fatigue property of a ferroelectric layer and maintain the switchable polarization of the ferroelectric material.

In the present disclosure, a device is disclosed. The device comprises: a gate stack and a channel layer over the gate stack. In some embodiments, the gate stack includes: a metal gate electrode, a ferroelectric layer, and a semiconducting oxide layer disposed between the ferroelectric layer and the metal gate electrode. In some embodiments, the semiconducting oxide layer has a thickness between approximately 1 μm and approximately 30 μm .

In the present disclosure, another device is disclosed. The device comprises a ferroelectric capacitor. In some embodiments, the ferroelectric capacitor includes: a first semiconducting oxide electrode layer, a second semiconducting oxide electrode layer, a ferroelectric layer between the first semiconducting oxide electrode layer and the second semiconducting oxide electrode layer, and a first semiconducting oxide layer disposed between the ferroelectric layer and the second semiconducting oxide electrode layer. In some embodiments, the first semiconducting oxide electrode layer has a thickness between approximately 1 μm and approximately 30 μm .

In the present disclosure, another device is disclosed. The device comprises a laminated structure. The laminated structure includes a ferroelectric layer, an electrode layer over the ferroelectric layer, and a semiconducting oxide electrode layer; wherein the ferroelectric layer is between the electrode layer and the semiconducting oxide electrode layer. In some embodiments, the semiconducting oxide electrode layer has a thickness between approximately 1 μm and approximately 30 μm .

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A device, comprising:
 - a gate stack including:
 - a metal gate electrode;

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- a ferroelectric layer, wherein the ferroelectric layer includes a ferroelectric material including HfSiO_x , HfZrO_x , Al_2O_3 , TiO_2 , LaO_x , BaSrTiO_x (BST), $\text{PbZr}_{1-x}\text{Ti}_x\text{O}_3$ (PZT), BiFeO_3 (BFO) or $(\text{Pb}_{1-x}\text{La}_x)(\text{Zr}_{1-x}\text{Ti}_x)\text{O}_3$ (PLZT), or a combination thereof;

- a semiconducting oxide layer disposed between the ferroelectric layer and the metal gate electrode, wherein the semiconducting oxide layer has a thickness between approximately 1 μm and approximately 30 μm ; and

- a channel layer over the gate stack.

2. The device of claim 1, wherein the semiconducting oxide layer includes SrRuO_3 , InGaZnO (IGZO) or LaSrMnO .

3. The device of claim 1, further comprising a high-k dielectric layer disposed between the ferroelectric layer and the channel layer.

4. The device of claim 1, wherein the ferroelectric layer is over the semiconducting oxide layer.

5. The device of claim 1, wherein the ferroelectric layer has a thickness between approximately 5 nm and approximately 20 nm.

6. A device, comprising:

- a ferroelectric capacitor, including:

- a first semiconducting oxide electrode layer,

- a second semiconducting oxide electrode layer;

- a ferroelectric layer between the first semiconducting oxide electrode layer and the second semiconducting oxide electrode layer, wherein the ferroelectric layer includes a ferroelectric material including HfSiO_x , HfZrO_x , Al_2O_3 , TiO_2 , LaO_x , BaSrTiO_x (BST), $\text{PbZr}_{1-x}\text{Ti}_x\text{O}_3$ (PZT), BiFeO_3 (BFO) or $(\text{Pb}_{1-x}\text{La}_x)(\text{Zr}_{1-x}\text{Ti}_x)\text{O}_3$ (PLZT), or a combination thereof; and

- a first semiconducting oxide layer disposed between the ferroelectric layer and the second semiconducting oxide electrode layer,

- wherein the first semiconducting oxide electrode layer has a thickness between approximately 1 μm and approximately 30 μm .

7. The device of claim 6, wherein the second semiconducting oxide electrode layer includes SrRuO_3 , InGaZnO (IGZO) or LaSrMnO .

8. The device of claim 6, wherein the second semiconducting oxide electrode layer has the same material as the first semiconducting oxide electrode layer.

9. The device of claim 8, wherein the first semiconducting oxide electrode layer and second semiconducting oxide electrode layer includes SrRuO_3 , InGaZnO (IGZO) or LaSrMnO .

10. The device of claim 6, wherein the first semiconducting oxide electrode layer includes SrRuO_3 , InGaZnO (IGZO) or LaSrMnO .

11. The device of claim 6, wherein the ferroelectric layer is separated from at least the first semiconducting oxide electrode layer or the second semiconducting oxide electrode layer.

12. The device of claim 6, wherein the ferroelectric layer has a thickness between approximately 5 nm and approximately 20 nm.

13. A device, comprising:

- a laminated structure including:

- a ferroelectric layer, wherein the ferroelectric layer includes a ferroelectric material including HfSiO_x , HfZrO_x , Al_2O_3 , TiO_2 , LaO_x , BaSrTiO_x (BST), $\text{PbZr}_{1-x}\text{Ti}_x\text{O}_3$ (PZT), BiFeO_3 (BFO) or $(\text{Pb}_{1-x}\text{La}_x)(\text{Zr}_{1-x}\text{Ti}_x)\text{O}_3$ (PLZT), or a combination thereof;

- an electrode layer over the ferroelectric layer;

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a semiconducting oxide electrode layer; wherein the ferroelectric layer is between the electrode layer and the semiconducting oxide electrode layer;

wherein the semiconducting oxide electrode layer has a thickness between approximately 1 μm and approximately 30 μm .

14. The device of claim **13**, wherein the semiconducting oxide electrode layer includes SrRuO_3 , InGaZnO (IGZO) or LaSrMnO .

15. The device of claim **13**, further including:
a channel layer; and
a source/drain electrode, wherein the channel layer is disposed between the source/drain electrode and the ferroelectric layer.

16. The device of claim **15**, further comprising a high-k dielectric layer disposed between the ferroelectric layer and the channel layer.

17. The device of claim **15**, wherein the source electrode and the drain electrode are separated by an insulating structure.

18. The device of claim **15**, further including a substrate and a buffer layer over the substrate, wherein the source electrode and the drain electrode are disposed over the buffer layer.

19. The device of claim **13**, wherein the ferroelectric layer is over the semiconducting oxide electrode layer.

20. The device of claim **13**, wherein the ferroelectric layer is separated from the electrode layer.

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